

Module 9: Outlier Detection

- ❑ Basic concepts
- ❑ Statistical approaches
- ❑ Proximity-based approaches
- ❑ Reconstruction-based approaches
- ❑ Clustering and classification based approaches
- ❑ Mining contextual and collective outliers
 - ❑ Transforming Contextual Outlier Detection to Conventional Outlier
 - ❑ Modeling Normal Behavior with Respect to Contexts
 - ❑ Mining Collective Outliers
- ❑ Outlier detection in high-dimensional data

Identifying Context

- ❑ Customer information attributes
 - ❑ Contextual attributes: age group (i.e., under 25, 25-45, 45-65, and over 65) and postal code
 - ❑ Behavioral attributes: number of transactions per year and annual total transaction amount
- ❑ Compare a customer c with the other customers in the same group, and use a conventional outlier detection method, such as some of the ones discussed earlier, to determine whether c is an outlier
- ❑ Learn a mixture models U on the contextual attributes and V on the behavior attributes
- ❑ The outlier score $S(o) = \sum_{U_j} \left(p(o \in U_j) \sum_{V_i} \left(p(o \in V_i) p(V_i | U_j) \right) \right)$
- ❑ If a context group is too small, assume that customers of similar age and who live within the same area should have similar normal behavior, then generalizing contexts based on age and location can be a reasonable first step for outlier detection.

Modeling Normal Behavior with Respect to Contexts

- ❑ In some applications, it is inconvenient or infeasible to clearly partition the data into contexts
 - ❑ Example: whether a customer purchases a product that is unrelated to the customer's recent interest – it is hard to define the context precisely
- ❑ Train a model that predicts the expected behavior attribute values with respect to the contextual attribute values
- ❑ To determine whether a data object is a contextual outlier, apply the model to the contextual attributes of the object – If the behavior attribute values of the object significantly deviate from the values predicted by the model, then the object can be declared a contextual outlier
 - ❑ Example models: Markov models and finite state automaton

Collective Outliers

- ❑ A group of data objects forms a collective outlier if the objects as a whole deviate significantly from the entire data set, even though each individual object in the group may not be an outlier
- ❑ To detect collective outliers, we have to examine the structure of the data set, that is, the relationships between multiple data objects
 - ❑ For outlier detection in temporal data (e.g., time series and sequences), explore the structures formed by time, which occur in segments of the time series or subsequences
 - ❑ To detect collective outliers in spatial data, explore local areas
 - ❑ In graph and network data, explore subgraphs
- ❑ The critical difference in collective outlier detection is that the structures are often not explicitly defined, and need to be discovered as part of the outlier detection process

Collective Outlier Detection on Graph Data

- ❑ Input: a social network as an unlabeled graph
- ❑ Treat each possible subgraph of the network as a structure unit
- ❑ For each subgraph S , extract two features
 - ❑ $|S|$: the number of vertices in S
 - ❑ $\text{freq}(S)$: the number of different subgraphs in the network that are isomorphic to S
- ❑ Outlier subgraphs
 - ❑ Small subgraphs that are of very low frequency
 - ❑ Large subgraphs that are surprisingly frequent